Slide 3: input and bias-T

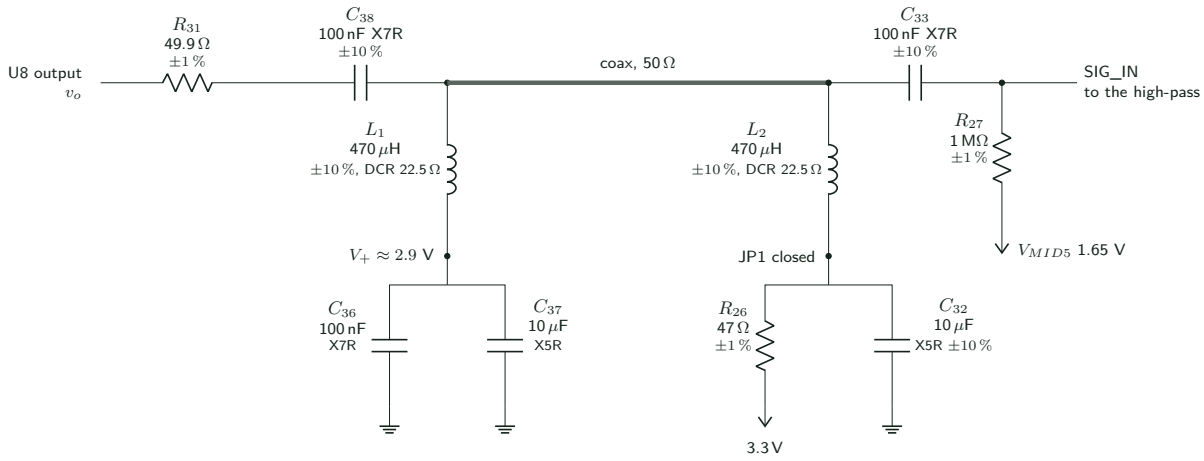
One coax carries the signal in and the power out

ADS7057 test board v3 — study notes

1 What the circuit must do

The sensor head has no battery. It needs about 4.3 mA of DC from the main board. The same coax must also bring the AE signal (100 to 500 kHz) back, without the supply wires loading it. A *bias-T* does this with two parts that behave in opposite ways:

- an **inductor** passes DC and blocks AC ($|Z_L| = 2\pi f L$),
- a **capacitor** blocks DC and passes AC ($|Z_C| = 1/2\pi f C$).



2 The DC path: how much voltage reaches the head

DC flows from +3.3 V through R26, the closed jumper JP1, L2, the coax centre, and L1 into the head. At DC the capacitors are open and the inductors are only their winding resistance, DCR = 22.5 Ω each (LCSC C75644). So

$$R_{\mathrm{DC}} = R_{26} + \mathrm{DCR}_{L2} + \mathrm{DCR}_{L1} = 47 + 22.5 + 22.5 = 92\,\Omega, \qquad \Delta V = I\,R_{\mathrm{DC}} = 4.3\,\mathrm{mA} \times 92\,\Omega = 0.40\,\mathrm{V},$$

$$V_+ = 3.3 - 0.40 = 2.90\,\mathrm{V}.$$

This is the supply of the OPA365. The head amplifier works from 2.2 to 5.5 V, so there is margin. The inductor is rated 40 mA, ten times the load.

R26 and C32 form a supply filter with a corner at $1/(2\pi \cdot 47 \cdot 10\,\mu\mathrm{F}) = 339$ Hz. It removes ripple from the 3.3 V rail before it enters the coax.

3 The AC path: what the signal sees

The head output drives R31 (49.9 Ω) and C38. Both inductors hang from the coax line to an AC ground (their far ends sit on large capacitors). Two branches $Z_b = R_{\mathrm{DCR}} + j\omega L$ are in parallel, so the line is loaded by

$$Z_L = \tfrac{1}{2}\,(R_{\mathrm{DCR}} + j\omega L).$$

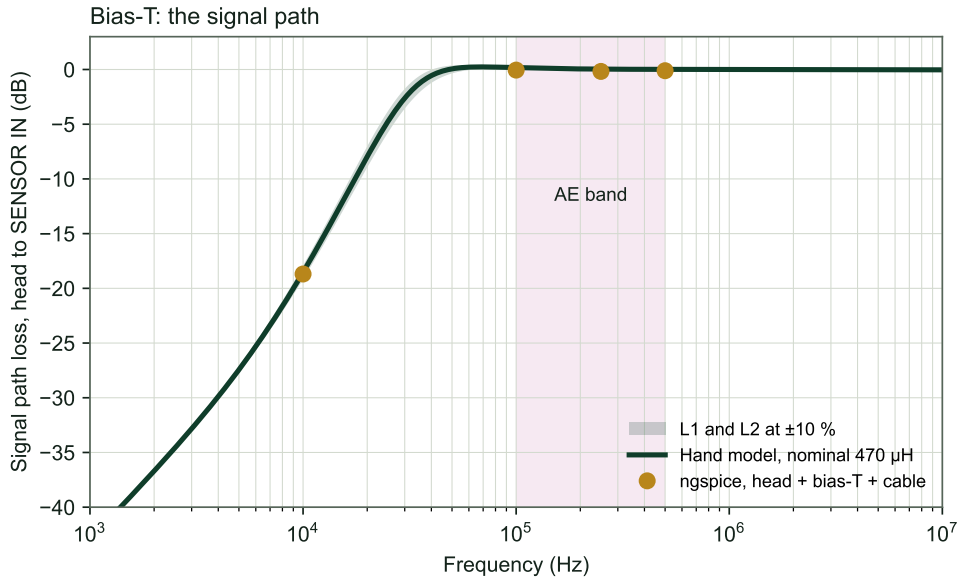
The source impedance is $Z_s = R_{31} + 1/(j\omega C_{38})$. The line voltage is a plain divider:

$$H(s) = \frac{Z_L}{Z_L + Z_s} = \frac{s^2\frac{LC}{2} + s\frac{RC}{2}}{s^2\frac{LC}{2} + s\,C\left(\frac{R}{2} + R_{31}\right) + 1}.$$

Read it as a **second-order high-pass** with

$$\omega_0 = \sqrt{\frac{2}{LC}}, \quad f_0 = \frac{1}{2\pi}\sqrt{\frac{2}{470\,\mu\mathrm{H}\cdot 100\,\mathrm{nF}}} = 32.8\,\mathrm{kHz}, \qquad Q = \frac{1}{\omega_0\,C\,(R/2 + R_{31})} \approx 0.79.$$

Below about 33 kHz the inductors short the line to ground, so the bias-T rejects low-frequency noise by itself. At 10 kHz the loss is 18.5 dB by hand and 18.7 dB in ngspice. At 250 kHz, the middle of the AE band, the hand model gives about 0 dB and ngspice gives 0.17 dB. The extra 0.1 dB comes from the cable and the filter loading, which the hand model leaves out.



4 Checks and tolerances

Item	Value	Effect
L1, L2: 470 µH ±10 %	f_0 moves ±5 % (31 to 34.5 kHz)	small, shown as the grey ban
Wavelength at 250 kHz	about 800 m in the cable	a 1 m cable is a lumped elem
Cable capacitance (RG174, about 100 pF/m)	6.4 kΩ at 250 kHz	adds a small load
Inductor self-resonance	2 MHz	above the band, so L stays a
C33 and R27 at SIG_IN	100 nF into 1 MΩ	1.6 Hz corner, DC blocked